

B.TECH ARTIFICIAL INTELLIGENCE | C05

Multi-Agent System *for Disaster Management*

Intelligent Agent Coordination for Real-Time Disaster Detection and Response

AI Course Technical Presentation

Vasanth Nithi D Y

192424137

SLIDE 2

Problem & Proposed Solution

The Problem



Speed

Disasters require **fast detection** and coordinated response — every second counts.



Data Overload

Massive real-time data streams from **sensors, satellites, weather systems** and emergency reports.

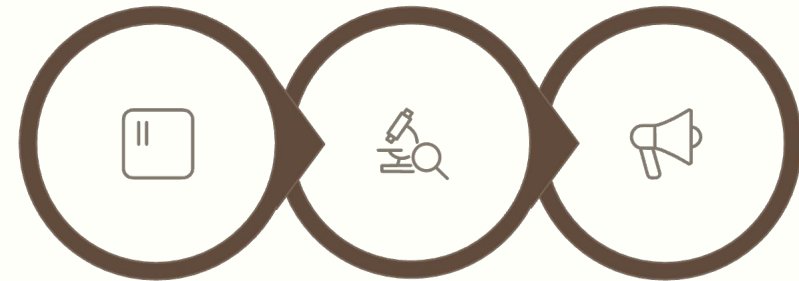


Human Limits

Human-only decision-making is **too slow** for rapidly evolving disaster scenarios.

Proposed Solution

A **Multi-Agent System (MAS)** where specialized AI agents collaborate, share information, and act autonomously to detect disasters and coordinate emergency response in real time.



Detect

Analyze

Respond

Multi-Agent Architecture



Sensor Agents

- Collect real-time data from IoT sensors, cameras, satellites, and weather systems
- Continuously stream environmental readings to the shared data pipeline
- Operate autonomously and in parallel across multiple geographic zones



Decision Agents

- Analyze incoming sensor data and assess disaster severity
- Combine information from multiple sources to determine appropriate actions
- Dynamically update decisions as the situation evolves



Communication Agents

- Send alerts to emergency services, authorities, and response teams
- Coordinate information flow between all stakeholders
- Distribute response instructions and situation updates in real time



Agents coordinate through a **shared data pipeline** and **distributed processing** — enabling parallel, scalable decision-making.

Coordination & Real-Time Processing



Continuous Monitoring

Sensor agents monitor environmental conditions — temperature, seismic activity, flood levels, and more — around the clock.



Distributed Stream Processing

Incoming data is processed in real time using distributed frameworks such as **Apache Spark / PySpark**, enabling high-throughput, low-latency analysis.



Multi-Source Data Fusion

Decision agents combine sensor streams, historical records, and satellite imagery to build a unified situational picture.



Dynamic Adaptation

Agents continuously share updates and **adapt decisions** as disaster conditions change, enabling real-time replanning.



Alert Distribution

Communication agents dispatch targeted alerts and response instructions to relevant emergency units immediately.



SLIDE 5

Benefits, Challenges & Conclusion

✓ Benefits

- Faster disaster detection
- Scalable processing of large real-time data
- Improved coordination across response units
- Real-time, data-driven decision-making
- Significantly reduced emergency response time

⚠ Challenges

Data Reliability

Sensor failures and noisy data can compromise situational awareness.

Communication Failures

Network disruptions may interrupt agent coordination during disasters.

Coordination Complexity

Managing consensus and conflict resolution among many agents is non-trivial.

Privacy & Security

Sensitive data from surveillance and sensors must be protected.

Conclusion: A Multi-Agent System provides a **scalable and intelligent** approach for disaster management — enabling specialized AI agents to collaborate, process real-time data, and support faster emergency response.